THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 858 cm²	TSA = CSA of cone + CSA of hemisphere $= \pi rl + 2\pi r^2$; TSA $= \frac{22}{7} \times 7 \times 25 + 2 \times \frac{22}{7} \times 7^2 = 858$ cm^2
2	**(b)** 44.2	$\sum f_i = 50$ and $\sum f_i x_i = 2210$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{2210}{50} = 44.2$
3	**(a)** $\frac{5}{36}$	Favourable outcomes: (2,6), (3,5), (4,4), (5,3), (6,2); number $= 5$; $P = \frac{5}{36}$
4	**(d)** 374 cm²	$2\pi rh + 2(2\pi r^2) = 2 \times \frac{22}{7} \times 3.5 \times 10 + 4 \times \frac{22}{7} \times 3.5^2$; TSA $= 374$ cm^2
5	**(d)** 78.75	Median class 70–80: $l = 70$, $cf = 9$, $f = 4$, $h = 10$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 70 + \frac{12.5 - 9}{4} \times 10 = 78.75$
6	**(a)** $\frac{3}{8}$	Favourable outcomes: HHT, HTH, THH; number $= 3$; $P = \frac{3}{8}$
7	**(d)** 1330 cm²	$6a^2 - \pi r^2 + 2\pi r^2 = 6a^2 + \pi r^2$ with $r = 7$ cm; TSA $= 1176 + 154 = 1330$ cm^2
8	**(c)** 72.5	$l = 70$, $f_1 = 13$, $f_0 = 10$, $f_2 = 4$, $h = 10$; Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 70 + \frac{3}{12} \times 10 = 72.5$
9	**(c)** $\frac{4}{11}$	Favourable letters: A, E, A, I; number $= 4$; $P = \frac{4}{11}$
10	**(c)** 1760 cm²	TSA = CSA cylinder + one flat base + CSA of cone; $= 1056 + 154 + 550 = 1760$ cm^2
11	**(d)** 30–40	It belongs to the class 30–40; Modal class = 30–40
12	**(c)** $\frac{1}{2}$	Favourable outcomes = number of red balls; $P = \frac{6}{12} = \frac{1}{2}$
13	**(c)** 374 cm²	$2\pi rh + 2(2\pi r^2)$ with $r = 3.5$, $h = 10$; TSA $= 220 + 154 = 374$ cm^2
14	**(d)** 16	$\frac{1650 + 35p}{34 + p} = 44.2$ gives $(35 - 44.2)p = -147.2$; $p = 16$
15	**(c)** $\frac{3}{10}$	Favourable numbers: 3, 6, 9, 12, 15, 18; number $= 6$; $P = \frac{6}{20} = \frac{3}{10}$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Volume of sphere = volume of wire (cylinder) $= \pi r^2 L$; $\frac{4}{3}\pi(6)^3 = \pi(2)^2 L$ (1); $L = 72$ cm (1).	1+1
17	Class marks x_i: 5, 15, 25, 35, 45, 55; $\sum f_i = 100$ and $\sum f_i x_i = 2620$ (1); Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{2620}{100} = 26.2$ (1).	1+1
18	Total outcomes $= 2^3 = 8$: HHH, HHT, HTH, HTT, THH, THT, TTH, TTT; Favourable outcomes: HHT, HTH, HTT, THH, THT, TTH, TTT; number $= 7$ (1); $P = \frac{7}{8}$ (1).	1+1
19	Volume of cuboid $= n\times$ volume of a cube; $n = \frac{24 \times 12 \times 9}{6^3}$ (1); $n = 12$ (1).	1+1
20	Highest frequency is 16, so the modal class is 80–90; $l = 80$, $f_1 = 16$, $f_0 = 13$, $f_2 = 7$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 80 + \frac{3}{12} \times 10 = 82.5$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Volume = hemisphere + cone (1); Hemisphere $= \frac{2}{3}\pi r^3 = 19404$ cm^3; cone $= \frac{1}{3}\pi r^2 h = 4620$ cm^3 (1); Volume $= 24024$ cm^3 (1).	1+1+1
22	Take assumed mean $a = 35$; deviations $d_i = x_i - a$: -30, -20, -10, 0, 10, 20 (1); $\sum f_i = 100$ and $\sum f_i d_i = -880$ (1); Mean $= a + \frac{\sum f_i d_i}{\sum f_i} = 35 + \frac{-880}{100} = 26.2$ (1).	1+1+1
23	Total number of outcomes $= 6 \times 6 = 36$ (1); Favourable outcomes: (1,1), (1,4), (2,2), (3,3), (4,1), (4,4), (5,5), (6,6); number $= 8$ (1); $P = \frac{8}{36} = \frac{2}{9}$ (1).	1+1+1
24	Volume = cylinder + two hemispheres $= \pi r^2 h + \frac{4}{3}\pi r^3$ (1); Cylinder $= 3465$ cm^3; two hemispheres $= 4851$ cm^3 (1); Volume $= 8316$ cm^3 (1).	1+1+1
	OR: Cumulative frequencies: 13, 28, 37, 45, 50; $N = 50$, so $\frac{N}{2} = 25$ (1); Median class 10–20: $l = 10$, $cf = 13$, $f = 15$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 10 + \frac{25 - 13}{15} \times 10 = 18$ (1).	1+1+1
25	Total possible outcomes $= 18$ (1); Favourable numbers: 3, 6, 9, 12, 15, 18; number $= 6$ (1); $P = \frac{6}{18} = \frac{1}{3}$ (1).	1+1+1
	OR: Volume left = cylinder − cone (1); Cylinder $= 1848$ cm^3; cone $= 616$ cm^3 (1); Volume left $= 1232$ cm^3 (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Volume of water is the same in both tanks (2); $\pi r^2 h = 1540$ cm^3 and base area of cuboid $= 77$ cm^2 (2); Height $= 1540 \div 77 = 20$ cm (1).	2+2+1
	OR: $\sum f_i = 21 + p$ and $\sum f_i x_i = 1535 + 75p$ (2); $\frac{1535+75p}{21+p} = 73.4$ gives $(75 - 73.4)p = 6.4$ (2); $p = 4$ (1).	2+2+1
27	Total outcomes $= 36$ (2); Favourable outcomes: 5; 6; 4 (2); Probabilities: $\frac{5}{36}, \frac{1}{6}, \frac{1}{9}$ (1).	2+2+1
	OR: Volume of water in the cone $= \frac{1}{3}\pi r^2 h$ (2); $\frac{1}{3}\pi(6)^2(8) = \pi(4)^2 x$ (2); $x = 6$ cm (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 32	Volume of a sphere $= \frac{4}{3}\pi(3)^3$; $n = \frac{3r^2h}{4\rho^3}$; $n = 32$
2	**(b)** 17.5	$\frac{10+25}{2}$
3	**(c)** $\frac{8}{9}$	Favourable outcomes = number of good pens; $P = \frac{120}{135} = \frac{8}{9}$
4	**(a)** 6 cm	$\frac{4}{3}\pi R^3 = \frac{1}{3}\pi(6)^2(24)$ gives $R^3 = 216$; $R = 6$ cm
5	**(c)** 23.8	$\sum f_i = 50$ and $\sum f_i x_i = 1190$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{1190}{50} = 23.8$
6	**(b)** $\frac{1}{6}$	Favourable outcomes: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1); number $= 6$; $P = \frac{6}{36} = \frac{1}{6}$
7	**(a)** 5082 cm²	TSA = CSA of cone + CSA of hemisphere $= \pi rl + 2\pi r^2$; TSA $= \frac{22}{7} \times 21 \times 35 + 2 \times \frac{22}{7} \times 21^2 = 5082$ cm²
8	**(d)** 65	Median class 60–70: $l = 60$, $cf = 19$, $f = 12$, $h = 10$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 60 + \frac{25-19}{12} \times 10 = 65$
9	**(c)** $\frac{1}{2}$	Favourable outcomes: HHH, HHT, HTH, THH; number $= 4$; $P = \frac{4}{8} = \frac{1}{2}$
10	**(d)** 1232 cm²	$2\pi rh + 2(2\pi r^2) = 2 \times \frac{22}{7} \times 7 \times 14 + 4 \times \frac{22}{7} \times 7^2$; TSA $= 1232$ cm²
11	**(d)** 85.5	$l = 80$, $f_1 = 16$, $f_0 = 5$, $f_2 = 7$, $h = 10$; Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 80 + \frac{11}{20} \times 10 = 85.5$
12	**(c)** $\frac{2}{11}$	Favourable letters: B, B; number $= 2$; $P = \frac{2}{11}$
13	**(d)** 5320 cm²	$6a^2 - \pi r^2 + 2\pi r^2 = 6a^2 + \pi r^2$ with $r = 14$ cm; TSA $= 4704 + 616 = 5320$ cm²
14	**(c)** 30–40	It belongs to the class 30–40; Modal class $=$ 30–40
15	**(a)** $\frac{2}{3}$	Favourable outcomes = number of red balls; $P = \frac{10}{15} = \frac{2}{3}$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Total number of outcomes $= 6 \times 6 = 36$; Favourable outcomes: (1,1), (1,2), (1,3), (2,1), (2,2), (3,1); number $= 6$ (1); $P = \frac{6}{36} = \frac{1}{6}$ (1).	1+1
17	Volume of water displaced = volume of sphere = rise × base area; $\frac{4}{3}\pi(3)^3 = \pi(3)^2 x$ (1); $x = 4$ cm (1).	1+1
18	Cumulative frequencies: 16, 38, 58, 63, 82, 100; $N = 100$, so $\frac{N}{2} = 50$; Median class 20–30: $l = 20$, $cf = 38$, $f = 20$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 20 + \frac{50-38}{20} \times 10 = 26$ (1).	1+1
19	Number of letters in the word $= 9$; Favourable letters: E, U, A, I, O; number $= 5$ (1); $P = \frac{5}{9}$ (1).	1+1
20	Volume of sphere = volume of wire (cylinder) $= \pi r^2 L$; $\frac{4}{3}\pi(6)^3 = \pi(3)^2 L$ (1); $L = 32$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Highest frequency is 20, so the modal class is 40–50 (1); $l = 40$, $f_1 = 20$, $f_0 = 16$, $f_2 = 4$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 40 + \frac{4}{20} \times 10 = 42$ (1).	1+1+1
22	After adding, red balls $= 3 + 3 = 6$ and total balls $= 15$ (1); Favourable outcomes = number of red balls (1); $P = \frac{6}{15} = \frac{2}{5}$ (1).	1+1+1
23	Volume = hemisphere + cone (1); Hemisphere $= \frac{2}{3}\pi r^3 = 19404$ cm³; cone $= \frac{1}{3}\pi r^2 h = 9240$ cm³ (1); Volume $= 28644$ cm³ (1).	1+1+1
24	Take assumed mean $a = 85$; deviations $d_i = x_i - a$: -20, -10, 0, 10, 20 (1); $\sum f_i = 40$ and $\sum f_i d_i = -360$ (1); Mean $= a + \frac{\sum f_i d_i}{\sum f_i} = 85 + \frac{-360}{40} = 76$ (1).	1+1+1
	OR: Total number of outcomes $= 6 \times 6 = 36$ (1); Favourable outcomes: (1,3), (2,2), (2,6), (3,1), (3,5), (4,4), (5,3), (6,2), (6,6); number $= 9$ (1); $P = \frac{9}{36} = \frac{1}{4}$ (1).	1+1+1
25	Volume = cylinder + two hemispheres $= \pi r^2 h + \frac{4}{3}\pi r^3$ (1); Cylinder $= 6930$ cm³; two hemispheres $= 4851$ cm³ (1); Volume $= 11781$ cm³ (1).	1+1+1

	OR: Cumulative frequencies: 2, 6, 8, 23, 25; $N = 25$, so $\frac{N}{2} = 12.5$ (1); Median class 40–50: $l = 40$, $cf = 8$, $f = 15$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 40 + \frac{12.5 - 8}{15} \times 10 = 43$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Median class is 30–40 with $f = 16$, and the cumulative frequency before it is $cf = 18 + x$ (2); $36.25 = 30 + \frac{50 - (18 + x)}{16} \times 10$ gives $x = 22$ (2); $x = 22$; then $y = 100 - 57 - x = 21$ (1).	2+2+1
	OR: $\frac{4 + x}{12 + x} = \frac{1}{2}$ (2); Cross-multiplying: $2(4 + x) = 1(12 + x)$ (2); $x = 4$ (1).	2+2+1
27	Volume of water is the same in both tanks (2); $\pi r^2 h = 1848$ cm^3 and base area of cuboid $= 308$ cm^2 (2); Height $= 1848 \div 308 = 6$ cm (1).	2+2+1
	OR: $\sum f_i = 38 + p$ and $\sum f_i x_i = 2540 + 65p$ (2); $\frac{2540 + 65p}{38 + p} = 66.4$ gives $(65 - 66.4)p = -16.8$ (2); $p = 12$ (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapters 12–14: **Surface Areas and Volumes, Statistics, Probability** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 7040 cm²	TSA = CSA cylinder + one flat base + CSA of cone; $= 4224 + 616 + 2200 = 7040$ cm^2
2	**(b)** 8	$\frac{990+25p}{42+p} = 23.8$ gives $(25 - 23.8)p = 9.6$; $p = 8$
3	**(a)** $\frac{1}{5}$	Favourable numbers: 1, 4, 9, 16, 25; number $= 5$; $P = \frac{5}{25} = \frac{1}{5}$
4	**(c)** 1276 cm²	$2\pi rh + 2(2\pi r^2)$ with $r = 7$, $h = 15$; TSA $= 660 + 616 = 1276$ cm^2
5	**(a)** $\frac{x_i - a}{h}$	Definition of u_i
6	**(a)** $\frac{8}{9}$	Favourable outcomes = number of good pens; $P = \frac{80}{90} = \frac{8}{9}$
7	**(c)** 81	Volume of a sphere $= \frac{4}{3}\pi(2)^3$; $n = \frac{3r^2h}{4\rho^3}$; $n = 81$
8	**(c)** 21.6	$\sum f_i = 50$ and $\sum f_i x_i = 1080$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{1080}{50} = 21.6$
9	**(c)** $\frac{1}{6}$	Favourable outcomes: (1,1), (2,2), (3,3), (4,4), (5,5), (6,6); number $= 6$; $P = \frac{6}{36} = \frac{1}{6}$
10	**(d)** 3 cm	$\frac{4}{3}\pi R^3 = \frac{1}{3}\pi(3)^2(12)$ gives $R^3 = 27$; $R = 3$ cm
11	**(d)** 34	Median class 30–40: $l = 30$, $cf = 21$, $f = 10$, $h = 10$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 30 + \frac{25-21}{10} \times 10 = 34$
12	**(a)** $\frac{1}{4}$	Favourable outcomes: HHH, TTT; number $= 2$; $P = \frac{2}{8} = \frac{1}{4}$
13	**(b)** 11770 cm²	TSA = CSA of cone + CSA of hemisphere $= \pi rl + 2\pi r^2$; TSA $= \frac{22}{7} \times 35 \times 37 + 2 \times \frac{22}{7} \times 35^2 = 11770$ cm^2
14	**(d)** 33.75	$l = 30$, $f_1 = 17$, $f_0 = 8$, $f_2 = 2$, $h = 10$; Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 30 + \frac{9}{24} \times 10 = 33.75$
15	**(a)** $\frac{5}{11}$	Favourable letters: X, M, N, T, N; number $= 5$; $P = \frac{5}{11}$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Class marks x_i: 65, 75, 85, 95, 105; $\sum f_i = 40$ and $\sum f_i x_i = 3040$ (1); Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{3040}{40} = 76$ (1).	1+1
17	Total outcomes $= 2^3 = 8$: HHH, HHT, HTH, HTT, THH, THT, TTH, TTT; Favourable outcomes: HHH, HHT, HTH, HTT; number $= 4$ (1); $P = \frac{4}{8} = \frac{1}{2}$ (1).	1+1
18	Volume of cuboid $= n\times$ volume of a cube; $n = \frac{24 \times 12 \times 15}{3^3}$ (1); $n = 160$ (1).	1+1
19	Highest frequency is 14, so the modal class is 70–80; $l = 70$, $f_1 = 14$, $f_0 = 10$, $f_2 = 2$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 70 + \frac{4}{16} \times 10 = 72.5$ (1).	1+1
20	Total number of outcomes $= 6 \times 6 = 36$; Favourable outcomes: (1,3), (2,4), (3,1), (3,5), (4,2), (4,6), (5,3), (6,4); number $= 8$ (1); $P = \frac{8}{36} = \frac{2}{9}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Total possible outcomes $= 28$ (1); Favourable numbers: 2, 3, 5, 7, 11, 13, 17, 19, 23; number $= 9$ (1); $P = \frac{9}{28}$ (1).	1+1+1
22	Volume left = cylinder − cone (1); Cylinder $= 3696$ cm^3; cone $= 1232$ cm^3 (1); Volume left $= 2464$ cm^3 (1).	1+1+1
23	Highest frequency is 16, so the modal class is 10–20 (1); $l = 10$, $f_1 = 16$, $f_0 = 8$, $f_2 = 8$, $h = 10$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 10 + \frac{8}{16} \times 10 = 15$ (1).	1+1+1
24	After adding, red balls $= 7 + 5 = 12$ and total balls $= 17$ (1); Favourable outcomes = number of red balls (1); $P = \frac{12}{17}$ (1).	1+1+1
	OR: Volume = hemisphere + cone (1); Hemisphere $= \frac{2}{3}\pi r^3 = 2425.5$ cm^3; cone $= \frac{1}{3}\pi r^2 h = 1617$ cm^3 (1); Volume $= 4042.5$ cm^3 (1).	1+1+1
25	Take assumed mean $a = 45$; deviations $d_i = x_i - a$: -20, -10, 0, 10, 20 (1); $\sum f_i = 40$ and $\sum f_i d_i = 10$ (1); Mean $= a + \frac{\sum f_i d_i}{\sum f_i} = 45 + \frac{10}{40} = 45.25$ (1).	1+1+1

	OR: Total number of outcomes $= 6 \times 6 = 36$ (1); Favourable outcomes: 15 pairs; number $= 15$ (1); $P = \frac{15}{36} = \frac{5}{12}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Total outcomes $= 36$ (2); Favourable outcomes: 6; 6; 6 (2); Probabilities: $\frac{1}{6}$, $\frac{1}{6}$, $\frac{1}{6}$ (1).	2+2+1
	OR: Volume of water in the cone $= \frac{1}{3}\pi r^2 h$ (2); $\frac{1}{3}\pi(6)^2(12) = \pi(3)^2 x$ (2); $x = 16$ cm (1).	2+2+1
27	Median class is 70–80 with $f = 16$, and the cumulative frequency before it is $cf = 44 + x$ (2); $71.25 = 70 + \frac{50-(44+x)}{16} \times 10$ gives $x = 4$ (2); $x = 4$; then $y = 100 - 76 - x = 20$ (1).	2+2+1
	OR: $\frac{5+x}{12+x} = \frac{8}{15}$ (2); Cross-multiplying: $15(5 + x) = 8(12 + x)$ (2); $x = 3$ (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND